



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

Name: ORDONEZ, ASHLEY NICOLE S.	Date: Sept. 25, 2026
Section: BSIT - 2B	Instructor: Ms. Maxyne Ignacio

- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer:

The program is designed to manage and organize IT automation incident tickets using a linear data structure. It allows the user to add, display, search, remove, and count incident tickets through a menu-driven system. The program helps simulate how support teams can manage active incident records in an organized and efficient way.

- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident:

```
tickets.append(ticket)
```

- The `append()` function adds a new incident ticket to the end of the tickets list. The user enters the Incident ID, Bot, and Short Description before the ticket is added.

Display Incidents:

for ticket in tickets:

- The for loop goes through each ticket stored in the tickets list and displays its Incident ID, Bot, and Short Description. This allows the user to view all active incident tickets.

Search Incident:

```
if ticket["incident_id"] == incident_id:
```

- The program compares the Incident ID entered by the user with the Incident ID of each ticket in the list. If a matching ID is found, the program displays the details of that incident ticket.



Remove Resolved Incident:

`tickets.remove(ticket)`

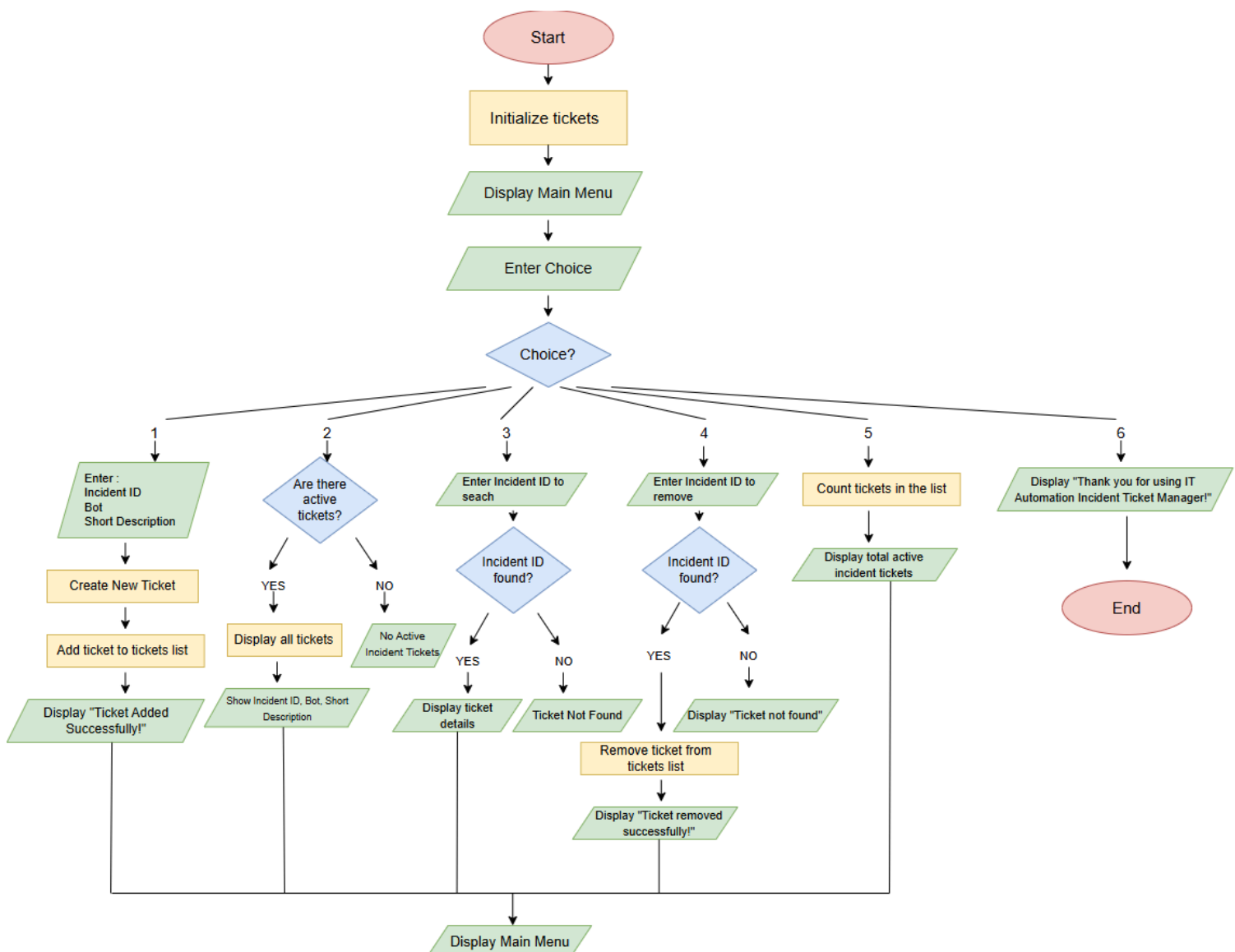
- The `remove()` function deletes the selected incident ticket from the tickets list when its Incident ID matches the ID entered by the user. This represents removing a resolved incident from the active tickets.

Count Active Incidents:

`len(tickets)`

- The `len()` function counts the number of tickets currently stored in the tickets list. It gives the total number of active incident tickets.

III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.





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- IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

Through this activity, I learned how a linear data structure can be used to store and manage records in a program. I learned how to perform basic operations such as adding, displaying, searching, removing, and counting incident tickets using a Python list. The activity also helped me understand how data structures can be applied to a practical IT-related scenario, such as managing automation incident tickets.